

Midterm 1

EECS 245, Fall 2026 at the University of Michigan

Name: _____

username: _____

UMID: _____

Room: ☐ 1013 DOW ☐ 1014 DOW (alternate/SSD) ☐ 3901 BBB (alternate)

Instructions

- This exam consists of 8 problems, worth a total of 100 points, spread across 14 pages (7 sheets of paper).
- You have 120 minutes to complete this exam, unless you have extended-time accommodations through SSD.
- Write your username in the top right corner of each page where a space is provided.
- When a large box is provided, **you must show all of your work and justify your thinking** there. Write your final answer in the small box, if one is provided. We will not grade work that appears outside the space provided. When justification is required, answers without sufficient justification will not receive full credit.
- For multiple choice problems, completely fill in bubbles and square boxes; if we cannot tell which option(s) you selected, you may lose points.
 - ☐ A bubble means that you should only select one choice.
 - ☐ A square box means that you should select all that apply.
- You may refer to **one double-sided 8.5x11" handwritten notes sheet**. Other than that, you may not refer to any other resources or technology during the exam (no phones, watches, or calculators).

You are to abide by the University of Michigan/Engineering Honor Code. To receive a grade, please sign below to signify that you have kept the Honor Code pledge.

I have neither given nor received aid on this exam, nor have I concealed any violations of the Honor Code.

This page has been intentionally left blank.

Tip: Skim through the entire exam before starting to work on it.

Problem 1 (12 pts)

Consider two datasets, $A = \{y_1, y_2, \dots, y_n\}$ and $B = \{z_1, z_2, \dots, z_m\}$. Let $R_A(w)$ and $R_B(w)$ represent the mean squared errors of the constant prediction w on datasets A and B , respectively:

$$R_A(w) = \frac{1}{n} \sum_{i=1}^n (y_i - w)^2 = (7 - w)^2 + 5, \quad R_B(w) = (13 - w)^2 + 17$$

a) (3 pts) Which dataset below could be A ?

- ☐ 7, 8, 10
 ☐ 4, 6, 8, 10
 ☐ 5, 6, 7, 8, 9
 ☐ 2, 3, 6, 7, 8

b) (3 pts) Let $S(w)$ be the **sum** of the two mean squared error functions:

$$S(w) = R_A(w) + R_B(w)$$

Find w^* , the value of w that minimizes $S(w)$. Give your answer as a number with no variables.

$w^* =$

c) (6 pts) Let $M(w)$ return the **larger** of the two mean squared error functions:

$$M(w) = \max(R_A(w), R_B(w))$$

Find w^* , the value of w that minimizes $M(w)$. Give your answer as a number with no variables. *Hint: Draw a picture — a clearly annotated picture, accompanied by some algebra, is sufficient justification.*

$w^* =$

Problem 2 (14 pts)

Consider a dataset of seven values $y_1, y_2, \dots, y_7$, listed in **non-decreasing** order:

$$y_1, y_2, 47, 60, 60, 81, y_7$$

Suppose that the mean of $y_1, \dots, y_6$ — that is, **not including** y_7 — is 50. Furthermore, suppose

- $f(w) = \frac{1}{7} \sum_{i=1}^7 |y_i - w|$, and α^* is the value of w that minimizes $f(w)$.
- $g(w) = \frac{1}{7} \sum_{i=1}^7 (y_i - w)^2$, and β^* is the value of w that minimizes $g(w)$.

a) (2 pts) What is the value of α^* ? Give your answer as a number with no variables.

$$\alpha^* = \boxed{}$$

b) (5 pts) What is the largest possible value of y_7 such that $\alpha^* \geq \beta^*$? Give your answer as a number with no variables.

largest possible $y_7 =$

username: _____

c) (7 pts) Consider the same seven values, listed in **non-decreasing** order:

$$y_1, y_2, 47, 60, 60, 81, y_7$$

Recall,

$$f(w) = \frac{1}{7} \sum_{i=1}^7 |y_i - w|$$

For this part, do not assume that $\alpha^* \geq \beta^*$.

Suppose that $f(59) = 30$. What is $f(65)$? Give your answer as a number with no variables.

Hint: Consider the slopes of $f(w)$ between $w = 59$ and $w = 65$.

$f(65) =$

Consider a dataset of n points, $(x_1, y_1), \dots, (x_n, y_n)$, where not all of the x_i are the same. We'd like to fit a **modified** simple linear regression model whose slope and intercept are forced to be the same:

The value of w that minimizes mean squared error for this model is

a) (3 pts) Suppose:

- Fill in the ??? with the relationship that is **guaranteed**:

$$0 > \quad 0 \geq \quad 0 = \quad 0 < \quad 0 \leq$$

- $$r = -\frac{3}{5}, \quad \sigma_x = 4, \quad \sigma_y = 10$$

(i) (3 pts) What is w^* , the optimal slope for the **modified** simple linear regression model?

(ii) (3 pts) What is w_1^* , the optimal slope for the **regular** simple linear regression model?

6

username: _____

- c) (7 pts) Show that w^* , the value of w that minimizes mean squared error for the modified simple linear regression model $h(x_i) = w + wx_i$, is

$$w^* = \frac{\sum_{i=1}^n (1 + x_i)y_i}{\sum_{i=1}^n (1 + x_i)^2}$$

Problem 4 (11 pts)

Suppose a , b , and c are real numbers, and let

$$\vec{u} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}, \quad \vec{v} = \begin{bmatrix} b \\ c \\ a \end{bmatrix}$$

- a) (5 pts) Use the Cauchy–Schwarz inequality to prove that the following holds for all real numbers a , b , and c :

$$|ab + bc + ca| \leq a^2 + b^2 + c^2$$

- b) (6 pts) Now suppose that $a + b + c = 0$ and $\vec{u} \neq \vec{0}$. Find θ , the angle between $\vec{u}$ and $\vec{v}$. Give your answer as the cosine inverse of a number (e.g. $\cos^{-1}(3/4)$). *Hint:* $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$.

$\theta = \cos^{-1}\left(\right.$ $\left.)\right)$

Problem 5 (12 pts)

Suppose $\vec{v}_1, \vec{v}_2 \in \mathbb{R}^2$ satisfy

$$\vec{v}_1 = \begin{bmatrix} 18 \\ -24 \end{bmatrix}, \quad \|\vec{v}_2\| = 6, \quad \vec{v}_1 \cdot \vec{v}_2 = 0$$

Let

$$\vec{x} = \begin{bmatrix} 10 \\ -5 \end{bmatrix}$$

- a) (5 pts) Find $\vec{p}$, the projection of $\vec{x}$ onto $\vec{v}_1$. Give your answer as a vector with two components and no variables.

$\vec{p} =$

- b) (7 pts) Suppose that the first component of $\vec{v}_2$ is positive. Write $\vec{x}$ as a linear combination of $\vec{v}_1$ and $\vec{v}_2$ by filling in the two boxes below.

$\vec{x} =$ $\vec{v}_1 +$ $\vec{v}_2$

Problem 6 (19 pts)

Throughout this problem, let

$$\vec{x}_1 = \begin{bmatrix} 12 \\ -2 \\ 6 \end{bmatrix}, \quad \vec{x}_2 = \begin{bmatrix} 3 \\ 0 \\ 3 \end{bmatrix}, \quad P = \text{span}(\{\vec{x}_1, \vec{x}_2\})$$

- a) (5 pts) Find an equation for the plane P of the form $Ax + By + Cz + d = 0$. Simplify your answer so that $A = 1$.

equation for plane:

For part **b)**, suppose that $\vec{x}_3, \vec{x}_4 \in \mathbb{R}^3$ and that $\text{span}(\{\vec{x}_1, \vec{x}_2, \vec{x}_3, \vec{x}_4\})$ is a **plane** in $\mathbb{R}^3$.

- b) (5 pts) For each statement below, determine whether it is impossible, possible (but **not** guaranteed), or guaranteed to be true, given the above assumptions. The first statement has been done for you as an example.

	statement	impossible?	possible?	guaranteed?
(i)	$\ \vec{x}_3\ = 5$.	☐	☒	☐
(ii)	$\vec{x}_3$ and $\vec{x}_4$ are scalar multiples of one another.	☐	☐	☐
(iii)	$\vec{x}_4 = \vec{0}$.	☐	☐	☐
(iv)	The vectors $\vec{x}_1, \vec{x}_2, \vec{x}_3$ are linearly independent.	☐	☐	☐
(v)	$\text{span}(\{\vec{x}_3, \vec{x}_4\}) = P$.	☐	☐	☐
(vi)	$\{\vec{x}_1, \vec{x}_2\}$ is a basis for P .	☐	☐	☐

Recall,

$$\vec{x}_1 = \begin{bmatrix} 12 \\ -2 \\ 6 \end{bmatrix}, \quad \vec{x}_2 = \begin{bmatrix} 3 \\ 0 \\ 3 \end{bmatrix}, \quad P = \text{span}(\{\vec{x}_1, \vec{x}_2\})$$

Additionally, suppose that $\vec{x}_3, \vec{x}_4 \in \mathbb{R}^3$ and that $\text{span}(\{\vec{x}_1, \vec{x}_2, \vec{x}_3, \vec{x}_4\})$ is a **plane** in $\mathbb{R}^3$.

- c) (4 pts) Give one nonzero vector orthogonal to $3\vec{x}_1 - 7\vec{x}_2 + 12\vec{x}_3 - \vec{x}_4$. Your answer should have no variables. Briefly justify it.

one vector =

For part d), suppose that $\vec{x}_3, \vec{x}_4 \in \mathbb{R}^3$ but that $\text{span}(\{\vec{x}_1, \vec{x}_2, \vec{x}_3, \vec{x}_4\}) = \mathbb{R}^3$.

- d) (5 pts) Which statements are **impossible**? Select all that apply.

- ☐ $\vec{x}_3$ and $\vec{x}_4$ are on P .
- ☐ The vectors $\vec{x}_3, \vec{x}_4$ are linearly independent.
- ☐ The vectors $\vec{x}_1, \vec{x}_2, \vec{x}_3, \vec{x}_4$ are linearly independent.
- ☐ There is a nonzero vector orthogonal to all four vectors.
- ☐ Every subset containing exactly three of the four vectors is a basis for $\mathbb{R}^3$.

Problem 7 (5 pts)

Consider the line L in $\mathbb{R}^3$ defined below.

$$L = \left\{ \begin{bmatrix} 6 \\ 2 \\ -4 \end{bmatrix} + t \begin{bmatrix} 3 \\ 1 \\ -2 \end{bmatrix} : t \in \mathbb{R} \right\}$$

a) (2 pts) True or False: L is a subspace of $\mathbb{R}^3$.

☐ True ☐ False

b) (3 pts) Consider the following two planes:

$$x + 2y + z = 0$$

$$4x + 4y + z = 0$$

Describe the set of points that lie on L and on **both** planes.

- ☐ There are no points in the intersection.
☐ A single point.
☐ An entire line of points.
☐ An entire plane of points.

Problem 8 (11 pts)

Consider the vectors $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_d \in \mathbb{R}^n$, where $d \geq 2$. For some scalar $t \in \mathbb{R}$, define:

$$\vec{s} = t(\vec{v}_1 + \dots + \vec{v}_d)$$

Then, for each $i = 1, 2, \dots, d$, define

$$\vec{u}_i = \vec{v}_i - \vec{s}$$

- a) (5 pts) For this part only, suppose $d = 3$. Find an expression for $\vec{u}_1 + \vec{u}_2 + \vec{u}_3$ in terms of t , $\vec{v}_1$, $\vec{v}_2$, and $\vec{v}_3$. Note that your answer **cannot** involve $\vec{s}$.

$\vec{u}_1 + \vec{u}_2 + \vec{u}_3 =$

- b) (4 pts) Find a value of t for which $\vec{u}_1, \dots, \vec{u}_d$ are **guaranteed** to be linearly dependent, regardless of the original vectors. Give your answer as an expression in terms of n , d , and/or constants. *Hint: Think about the formal definition of linear independence.*

$t =$

- c) (2 pts) Suppose $t > 0$. Fill in the ??? with the relationship that is **guaranteed**:

$$t(\|\vec{v}_1\| + \|\vec{v}_2\| + \dots + \|\vec{v}_d\|) \quad \text{???} \quad \|\vec{s}\|$$

☐ $>$
☐ $\geq$
☐ $=$
☐ $<$
☐ $\leq$

Congrats on finishing Midterm 1!

Feel free to draw us a picture about EECS 245 in the box below (or use it for scratch work).



Did you notice any violations of the Honor Code during the exam? If so, share details with us here. We will keep your identity anonymous when investigating any cases.

